

Game 04 — Word Puzzle (Complete Plan)

Game 4 of 7 Complexity: ★★ Medium Estimate: 2–3 days Status: Built

Complete plan (supersedes the 2026-09-09 sample). Status: **Built** (P0–P2 complete, 2026-09-11) — full game in `apps/frontend/public/games/word-puzzle/` (`core.js/game.js/style.css` + `data/themes.json`), jest suite + `core.test.html` green (incl. the 100-seeds-per-level generator proof and data lint), browser smoke test covered drag / tap-tap / keyboard paths, hints, stars, progress and the theme-complete flow; P3 extras (daily puzzle, confetti, more themes) deferred. Interpretation decided: **word search** (master README §8 #1–2). Slug `word-puzzle`; folder `apps/frontend/public/games/word-puzzle/`.

1. Overview

Themed word search: drag a line through letters to find hidden words; correct finds lock in with colored highlights, wrong picks flash red. Hook: 3-star levels and the daily seed (P3) that gives everyone the same puzzle.

2. Complete game spec

- **Content:** 4 themes (Animals 🐾, Food 🍎, Tech 💻, Space 🚀) × 3 levels, shipped as static `data/themes.json` (schema §5). English, lowercase, 4–9 letters, no spaces, no duplicate words within a level.
- **Grid tiers:** L1 = 8×8, 5 words, directions {→, ↓}; L2 = 10×10, 7 words,
 - {↖, ↗}; L3 = 12×12, 10 words, all 8 directions incl. reversed.
- **Selection:** pointerdown on a cell anchors; pointermove samples cells via `document.elementFromPoint`, snapped to the grid; a valid drag follows one of the 8 direction vectors in a straight line. Live highlight while dragging; on release:
 - Letters match an unfound word (or its reverse where reversed dirs are on) → **locked** in the word's color, struck off the list, +pop animation. ✅
 - Otherwise → red flash 300 ms, **wrong picks +1**, selection clears. ❌
- **Hints:** 3 per level; a hint rings the first letter of a random unfound word.
- **Stars:** ★ finish the level · ★ time ≤ par · ★ `hintsUsed === 0 && wrongPicks ≤ 2`.
- **Par times:** L1 90 s · L2 150 s · L3 240 s (per level data, tunable).
- **Level flow:** complete → result overlay → next level → theme complete → badge. Progress persisted per level (§5).

3. Generator (pure, deterministic)

`generateLevel(levelData, seed)` using `mulberry32(seed)` :

1. Sort words longest-first. For each word: try up to 200 random placements (random direction from the tier's set, random start cell) — legal if in bounds and every cell is empty or holds the same letter (shared-letter crossings allowed).
2. If a word fails 200 attempts → restart the whole level (max 20 restarts, then throw — data bug, must never ship).
3. Fill empty cells from a letter bag: 60 % weighted toward the level's own letters, 40 % uniform A–Z (keeps grids solvable-feeling and dense).
4. Return `{ grid, placements: [{word, row, col, dir}], seed }`.

4. Screens & UI

State	Elements
menu	Theme cards (emoji, name, stars x/9, badge when complete), Continue banner, mute toggle
level	Letter grid + word list (chips; found = struck, colored), HUD: timer, hint stars, [Hint] [△]
result	Overlay: time, stars, words found, [Next level] [Replay] [🔖 Share]; theme-complete variant

Word list sits below the grid at phone width (side-by-side ≥ 768 px). Grid cells sized `min(92vw, 100% / size)`; touch highlight uses per-word colors from a fixed 10-color accessible palette (4.5:1 on white).

5. Data model

```
data/themes.json:
{ "themes": [ { "id": "animals", "name": "Animals", "emoji": "🐾",
  "levels": [ { "size": 8, "parSec": 90, "words": ["LION","ZEBRA", ...] }, ... ] }, ... ] }

localStorage:
game:word-puzzle:progress → { "<themeId>:<lvl>": { stars: 0-3, bestTimeMs: 84200 } }
game:word-puzzle:prefs    → { muted: false }
```

6. File structure & function inventory

```
word-puzzle/
  index.html
  style.css      # grid, highlight pills, flash animations
  core.js        # generator + validation (pure, seeded) – test surface
  game.js        # input, rendering, timer, stars, persistence
  data/themes.json
```

`core.js` exports: `mulberry32(seed)`, `generateLevel(level, seed)`, `lineCells(grid, start, end) → cells | null` (8-direction check), `lettersAt(grid, cells) → string``,

`matchesWord(letters, word, allowReverse)`, ``starsFor(parSec, timeSec, hintsUsed, wrongPicks)` . `game.js`: pointer/tap-tap/keyboard input, highlight rendering, timer, progress, share.

7. Edge cases

1. Diagonal reverse drags (e.g. ✓ when reversed on) must validate as the word reversed.
2. Drag that leaves the grid → clamp to last valid cell; release outside → evaluate.
3. Re-finding a found word → counts as correct-but-noop (no star penalty, no duplicate).
4. Wrong selection on cells that partially spell a word (prefix) → still wrong (full word required) — documented in-game via the red flash.
5. Tap-tap mode: first tap anchors (cell pulses), second tap submits; tapping the anchor again cancels.
6. Keyboard: arrows move focus; Enter anchors/submits; grid is one tab stop (roving focus), `aria-label` per cell = "row r column c, letter X"; word chips labeled "word k, N letters" (never the word itself).
7. `visibilitychange` pauses the timer; level restarts only via user action.
8. Storage disabled → progress not saved, everything else works.

8. Testing plan (core.js)

- Seeded determinism: same seed → identical grid/placements (snapshot).
- Every word present in `placements`, within the tier's allowed directions.
- Generated grid: all placements verified by `lineCells` + `lettersAt` round-trip.
- Generator termination: 100 seeds per shipped level, zero throws.
- `lineCells`: 8 directions, out-of-bounds, non-collinear → null.
- `matchesWord`: exact, reverse (allowed vs not), case handling.
- `starsFor`: boundary table (par ±1 s, hints 0/1, wrong 2/3).

9. Task breakdown

P0 — playable core

- ✓ `core.js`: mulberry32, generateLevel, lineCells, lettersAt, matchesWord + tests
- ✓ Grid rendering + drag selection with live highlight and correct/wrong resolution
- ✓ Word list check-off, level timer, win detection, next-level flow

P1 — full rules & feel

- ✓ 4 themes × 3 levels data (`themes.json`) + data lint (lengths, dupes, charset)
- ✓ Reversed directions at L3; hints; star rating; color pills; red flash; pop anim
- ✓ Progress persistence + Continue; sound blips + mute

P2 — persistence/share/QA

- ✓ Tap-tap + full keyboard path; a11y labels (\$7.5–7.6)

- ☒ Share text (`{words} words in m:ss · ★★`); QA gate (master README §5) (browser smoke test at 390×844 covered seeded drags, tap-tap branches, keyboard, hints, stars, persistence, theme completion + lint harness; a human 10-minute session is still owed before calling it player-proof)

P3 — polish

- ☐ Daily puzzle (seed = `YYYYMMDD` hash → same grid for everyone, share-friendly); confetti on theme completion; more themes `_(deferred — ?seed=` QA hook already ships the deterministic-generator half)_

10. Acceptance criteria

- §8 tests green; every shipped level solvable and every word placeable (generator proof).
- Wrong picks never permanently mark letters; found letters stay locked.
- Drag usable at 360 px without zoom; grid + chips visible during a drag.
- Seeded daily mode reproduces identical grids across reloads.

11. Deferred coupling (intentionally NOT built)

Riddle/content-pipeline integration (backend word lists), footer/nav links, analytics events, achievements — per master README §7.